

Ritam Chakraborty

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science, Data Science

GPA: 3.87 • Berkeley, CA

Expected Graduation: 05/2028

- Relevant Coursework: Algorithms, Data Structures, Machine Learning, Optimization, Data Science, Probability & Random Processes, Discrete Math & Probability, Statistics, Abstract Linear Algebra, Multivariable Calculus, Numerical Analysis

EXPERIENCE

Software Engineering Intern

Capital One

Richmond, VA

06/2026 – 08/2026

- Designed and delivered an automated recovery framework using AWS EventBridge, SQS with DLQ, and Lambda to resolve recurring receipt drops caused by transient partner API outages, achieving an 88% reduction in manual on-call interventions
- Shipped an idempotent retry architecture with check-before-write state validation and micro-batching, eliminating duplicate receipt uploads and Lambda timeout failures across high-volume production traffic
- Wrote end-to-end integration tests covering retry success paths, DLQ routing, and duplicate-record handling, catching 3 edge-case failures in code review before they reached production

Monetization & Analytics Lead

Berkeleytime

Berkeley, CA

09/2025 – Present

- Direct a 5-engineer team building backend services (Express.js, MongoDB, Redis, GraphQL) for Berkeleytime, a course discovery platform serving over 10,000 UC Berkeley students; own architecture decisions and review all PRs merged to main
- Designed and shipped a threshold-based course alert system with background job workers, cutting missed enrollment notifications by over 80% and reducing alert latency to under one minute
- Built a user activity scoring model with configurable recency- and engagement-based decay, and a dashboard surfacing it, which the team used to reprioritize two roadmap features toward higher-usage flows

Full Stack Software Engineer (Contract)

Impact Observatory

Remote

02/2026 – 04/2026

- Built core architecture for a geospatial analytics platform end-to-end — React/TypeScript frontend, Python FastAPI backend, and a LangGraph-based LLM orchestration layer integrated with OpenAI — while coordinating a 6-person contractor team
- Designed an LLM agent tooling pipeline using LangGraph with RAG over 100+ product documents for intent routing
- Built a natural language query layer that converts user queries into GeoJSON boundaries for geospatial search and filtering

Software Engineering Intern

SomeIdea AI

Berkeley, CA

03/2025 – 12/2025

- Built an NLP-powered transcript analysis platform in Python/FastAPI that condensed 100+ pages of financial documents into actionable summaries, cutting manual research time for 50+ paid users by over 90%
- Migrated 1,000+ documents from Google Cloud Firestore to Amazon RDS, designing normalized relational schemas and Python/SQL ETL pipelines that cut data retrieval time by 60% under increased load

PROJECTS

vigil • Rust, Tokio

- Built a Rust log analytics engine backed by an async Tokio daemon that ingests JSON event streams and maintains separate in-memory field and time indexes, serving sub-millisecond filtering and aggregation to concurrent clients over a Unix socket
- Supported latency and error-rate analysis across millions of events through a custom query language and hand-written parser with boolean predicates, time-range filters, and streaming aggregations including counts, averages, and p50/p99 percentiles
- Reduced daemon restart time by 1.6–1.8x (up to 840ms at 1M events) with a checksummed write-ahead log and periodic snapshots, so recovery replays only records newer than the last snapshot instead of re-parsing the entire log from disk
- Validated recovery gains with a 600-run statistical benchmark suite (Welch's t-tests, 95% confidence intervals) across graceful and crash restarts, uncovering and fixing two durability bugs that silently blocked snapshot recovery above 47K events

pgraph • Go

- Created a probabilistic graph inference engine in Go that models directed graphs with independent Bernoulli edge probabilities, enabling exact and sampled reachability analysis and probabilistic path queries for risk and reliability use cases
- Optimized core probabilistic graph algorithms, including max-likelihood path search via a modified Dijkstra's algorithm, top-K path enumeration with Yen's algorithm, and concurrent Monte Carlo reachability simulation using goroutines
- Designed a custom query DSL with JSON graph and result serialization for expressive, composable probabilistic queries

SKILLS

Languages: Rust, Python, Go, C++, Java, SQL, TypeScript

Backend: FastAPI, Flask, Spring, Node.js, Express.js

Systems & Infrastructure: Docker, Kubernetes, AWS, Linux, Git

Data & ML: Pandas, NumPy, SciPy, Matplotlib, PyTorch, TensorFlow

Databases: PostgreSQL, MongoDB, DynamoDB